

Lecture 16, May 01 2026.

New Taylor Series From old.

1). Geometric Model:

$$\frac{1}{1-\Delta} = \sum_{n=0}^{\infty} \Delta^n \quad ; \quad |\Delta| < 1.$$

2). arctangent model:

$$\arctan \Delta = \sum_{n=0}^{\infty} \frac{(-1)^n \Delta^{2n+1}}{2n+1} \quad ; \quad |\Delta| < 1.$$

3). Sine Model:

$$\sin(\Delta) = \sum_{n=0}^{\infty} \frac{(-1)^n \Delta^{2n+1}}{(2n+1)!} \quad ; \quad \Delta \in (-\infty, \infty)$$

4). Cosine Model:

$$\cos(\Delta) = \sum_{n=0}^{\infty} \frac{(-1)^n \Delta^{2n}}{(2n)!} \quad ; \quad \Delta \in (-\infty, \infty).$$

Example: Find the Maclaurin series for:

a). $f(x) = x \cos x$.

b). $f(x) = \ln(1+3x^2)$.

Sln: a).

$$x \cos x = x \cdot \left(\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \right) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n)!} \quad ; \quad x \in (-\infty, \infty)$$

b). Find a Model for $\ln(1+\Delta)$.

$$\frac{d}{dx} (\ln(1+x)) = \frac{1}{1+x} = \frac{1}{1 - \underbrace{(-x)}} = \sum_{n=0}^{\infty} (-1)^n x^n$$

$$\text{So, } \ln(1+x) = \int \left(\sum_{n=0}^{\infty} (-1)^n x^n \right) dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1} + C.$$

if $x=0$ then

$$\ln(1+0) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} (0)^{n+1} + C \Rightarrow C=0.$$

$$5). \quad \ln(1+\Delta) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} \Delta^{n+1} \quad ; \quad |\Delta| < 1.$$

Now,

$$\ln(1 + \underbrace{3x^2}_{\Delta}) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} (\underbrace{3x^2}_{\Delta})^{n+1}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} 3^{n+1} x^{2n+2}$$

Finding the radius; $|\Delta| < 1 \Leftrightarrow |3x^2| < 1$

$$x^2 < \frac{1}{3}$$

$$\sqrt{x^2} < \frac{1}{\sqrt{3}}$$

$$|x| < \frac{1}{\sqrt{3}} = R$$

Example. Find the function represented by the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n 2^n x^n}{n!}$$

Sln: 6). The e^x model:

$$e^{\Delta} = \sum_{n=0}^{\infty} \frac{\Delta^n}{n!} ; \Delta \in (-\infty, \infty)$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n 2^n x^n}{n!} = \sum_{n=0}^{\infty} \frac{(\underbrace{-2x}_{\Delta})^n}{n!} = e^{-2x}$$

Example. Find the sum of the series

$$\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 2^2} + \frac{1}{3 \cdot 2^3} - \frac{1}{4 \cdot 2^4} + \dots$$

Sln:

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n \cdot 2^n} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \cdot \left(\frac{1}{2}\right)^n$$

$$\ln(1+\Delta) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} \Delta^{n+1} = \sum_{\substack{k=1 \\ k=n+1}}^{\infty} \frac{(-1)^{k+1}}{k} \Delta^k = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \Delta^n$$

So

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \left(\frac{1}{2}\right)^n = \ln\left(1 + \frac{1}{2}\right) = \ln\left(\frac{3}{2}\right)$$

Example. a). Evaluate $\int e^{-x^2} dx$ as an infinite series.

b). Evaluate $\int_0^1 e^{-x^2} dx$ correct to within an error of 0.001.

Sln: a). Finding the Maclaurin series for $f(x) = e^{-x^2}$ we have:

$$e^{-x^2} = e^{\Delta} = \sum_{n=0}^{\infty} \frac{\Delta^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!} ; \quad x \in (-\infty, \infty).$$

$\Delta = -x^2$

Now,

$$\int e^{-x^2} dx = \int \left(\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!} \right) dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)n!} x^{2n+1} + C$$

b). The Fundamental Theorem of Calculus gives:

$$\begin{aligned} \int_0^1 e^{-x^2} dx &= \left[x - \frac{x^3}{3 \cdot 1!} + \frac{x^5}{5 \cdot 2!} - \frac{x^7}{7 \cdot 3!} + \frac{x^9}{9 \cdot 4!} - \dots \right]_0^1 \\ &= 1 - \frac{1}{3 \cdot 1!} + \frac{1}{5 \cdot 2!} - \frac{1}{7 \cdot 3!} + \frac{1}{9 \cdot 4!} - \dots \\ &\approx 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} \approx 0.7475. \end{aligned}$$

Example. Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.

Sln:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

So,

$$e^x - 1 - x = \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

then,

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} &= \lim_{x \rightarrow 0} \frac{\left(\frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \right)}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{\cancel{x^2} \left(\frac{1}{2!} + \frac{x}{3!} + \frac{x^2}{4!} + \dots \right)}{\cancel{x^2}} \\ &= \frac{1}{2!} = \frac{1}{2}. \end{aligned}$$